

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Engineering Technology, Grade 12

University/College Preparation

TMJ4M

This course enables students to further develop knowledge and skills related to design, process planning, control systems, project management, quality assurance, and business operations. Students will use a broad range of tools and equipment, enhance their skills in computer-aided design, and collaborate in managing a project. Students will critically analyse and solve complex problems involved in manufacturing products. Students will expand their awareness of environmental and societal issues and of career opportunities in the manufacturing industry.

Prerequisite: Manufacturing Engineering Technology, Grade 11,
University/College Preparation

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the business operations associated with manufacturing and explain their role in product development;
- A2.** demonstrate an understanding of how to optimize individual or mass production systems by improving material flow, factory design, product layout, labour productivity, and quality control;
- A3.** demonstrate a working knowledge of various types of control systems used in the manufacturing process;
- A4.** apply relevant mathematical skills, scientific concepts, and technological literacy and communication skills in planning and implementing manufacturing processes.

SPECIFIC EXPECTATIONS

A1. Manufacturing Business Operations

By the end of this course, students will:

- A1.1** explain the titles and roles shown on a standard manufacturing organization chart (*e.g., president, controller, purchasing/inventory control, human resources, marketing, engineering, sales, production*);
- A1.2** describe and explain activities associated with product development in manufacturing (*e.g., product research, product testing, product improvement*);
- A1.3** describe activities associated with marketing manufactured products (*e.g., market research, promotion, sales, distribution*);
- A1.4** identify factors to be considered in estimating the cost of manufacturing a product (*e.g., labour and materials, capital equipment, process costs, location/transportation*) and explain their importance.

A2. Process Planning

By the end of this course, students will:

- A2.1** describe on the basis of research how to optimize production (*e.g., in terms of material flow, production layout, quality control, facility layout*) in various types of manufacturing facilities and systems;

- A2.2** demonstrate an understanding of factors associated with labour costs in manufacturing a product (*e.g., labour-intensive versus capital-intensive production methods; use of unskilled, semi-skilled, and skilled labour*);

- A2.3** explain the technological systems approach to a manufacturing enterprise (*e.g., with respect to people, knowledge, materials, energy, finance, capital*);

- A2.4** plan in collaboration with others a detailed manufacturing process for production (*e.g., a process plan for part routing, a process plan that includes the use of robotics*).

A3. Control Systems

By the end of this course, students will:

- A3.1** explain the use of a variety of control systems (*e.g., electronic, pneumatic, hydraulic, mechanical*) to automate processes;
- A3.2** design a manufacturing system (*e.g., robotic arm, drive system, lifting device*) using a variety of electronic, pneumatic, hydraulic, and mechanical control systems;
- A3.3** select appropriate control systems for project production, and calculate power requirements (*e.g., load requirements, horsepower*).

A4. Mathematics, Science, and Technological Literacy

By the end of this course, students will:

A4.1 apply advanced mathematical skills in performing a variety of tasks required within the context of manufacturing design and production (*e.g., calculate machine feeds and speeds, estimate costs, calculate ratios, calculate angles using trigonometry*);

A4.2 apply scientific concepts to manufacturing-related processes (*e.g., properties of materials for selection purposes, work energy theorem for energy consumption measurements, non-destructive material testing to measure material quality and safety*);

A4.3 use appropriate technical language and forms of communication related to the design, fabrication, and control of a product (*e.g., when preparing and explaining flow charts, operations and inspections charts, job descriptions, lists of tooling requirements, quality-control program materials; when describing certification requirements and specialized training; when making presentations; when completing a bill of material*).

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process to plan and develop solutions, products, or services in response to challenges in manufacturing technology;
- B2.** demonstrate an understanding of the management of a manufacturing enterprise and the interrelationships among its major areas of activity such as marketing, cost control, quality assurance, production, and inventory control;
- B3.** demonstrate the safe and effective use of tools, equipment, and materials in the production of a product or the development of a production process;
- B4.** develop and use a quality assurance system to industry standards in the production of a project.

SPECIFIC EXPECTATIONS

B1. Design and Planning Process

By the end of this course, students will:

- B1.1** use reverse engineering to explain existing products or processes in terms of function;
- B1.2** demonstrate proficiency in using a design process to plan and develop solutions to manufacturing-related challenges;
- B1.3** create a working drawing (*e.g., orthographic, isometric, pictorial*) with the use of computer-aided design that includes appropriate information (*e.g., geometric dimensioning and tolerancing, section views, symbols and abbreviations*) to assist in the development of solutions to manufacturing challenges;
- B1.4** use a variety of communication techniques (*e.g., multimedia presentation, electronic presentation*) and supporting documentation (*e.g., prints, technical reports, statistical charting*) to present and explain a process design and plan;
- B1.5** generate product specifications through the accurate interpretation of engineering drawings, sketches, and reports;
- B1.6** select suitable materials for fabricating products based on the design specifications and the intended use of the products;
- B1.7** use a design and planning process to solve a manufacturing challenge (*e.g., design and build a robot or control system prototype; design an auto-*

mated manufacturing system using process control charts; design solutions to create an ergonomically effective work environment).

B2. Project Management

By the end of this course, students will:

- B2.1** demonstrate an understanding of management roles in the development of a product (*e.g., planning: setting goals to establish course of action; organizing: structuring the job into manageable tasks; directing: assigning tasks and supervising their completion; controlling: comparing results against the original plan*);
- B2.2** demonstrate an understanding of the management of a manufacturing enterprise (*e.g., set up and manage, in collaboration with others, a small-scale manufacturing enterprise; visit and document the activities of a local manufacturing enterprise*) and its major areas of activity (*e.g., research and development, production, marketing, and finance*);
- B2.3** create, in collaboration with others, a detailed process plan (*e.g., including robotic applications and other tools, machines, and equipment required*) to fabricate a final product;
- B2.4** create, in collaboration with others, a product or process prototype/model (*e.g., rapid prototyping process*);

- B2.5** analyse the results of a production process, and modify operations, systems, and tooling as necessary.

B3. Tools, Equipment, and Manufacturing Processes

By the end of this course, students will:

- B3.1** demonstrate the skills required to safely operate machine tools and equipment (*e.g., engine lathe, milling machine, drill press; equipment for gas tungsten arc welding [GTAW] and gas metal arc welding [GMAW]*) in the assembly or fabrication of a product;
- B3.2** use computers to operate and control systems (*e.g., for inventory, quality control, computer numerical control [CNC], programmable logic control [PLC]*) in the assembly or fabrication of a product;
- B3.3** assemble and use power control and automation systems to meet design and process planning criteria;
- B3.4** demonstrate the safe and proper use of advanced cutting processes (*e.g., plasma, laser, water jet*);
- B3.5** demonstrate the appropriate use of advanced layout and set-up tools (*e.g., sine bar, gauge blocks, bevel protractor*);
- B3.6** use advanced measuring tools (*e.g., precision level, coordinate measuring machine, laser levels, optical comparators*) to calculate measurements in both metric and US customary / British imperial units;
- B3.7** operate computer numerical control equipment (*e.g., CNC milling machine, CNC lathe, CNC plasma cutter*) in the assembly or fabrication of a product.

B4. Quality Assurance

By the end of this course, students will:

- B4.1** identify organizations that develop standards regarding the conception and manufacture of consumer goods (*e.g., Canadian Standards Association [CSA], International Organization for Standardization [ISO]*);
- B4.2** demonstrate a working knowledge of how quality assurance is used to maintain design specifications (*e.g., through quality inspection and testing procedures*);
- B4.3** manage and control quality using industry-standard techniques and processes (*e.g., statistical process control [SPC], total quality management [TQM]*);
- B4.4** develop products and processes to meet customer specifications and quality control standards.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of ways in which the manufacturing industry affects the environment, and make informed decisions based on this understanding;
- C2.** assess the impact of the globalization of the manufacturing industry on society at the local, provincial, and national levels.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** identify potentially harmful consequences of manufacturing activities for the environment (*e.g., waste disposal, greenhouse gas emissions, water and energy consumption, the depletion of non-renewable resources*), and formulate alternatives to reduce the severity of these consequences;
- C1.2** assess and compare energy sources (*e.g., renewable – water, wind, solar, geothermal; non-renewable – coal, oil and gas, nuclear*) used in manufacturing, and identify ways of increasing environmentally friendly energy use;
- C1.3** assess the carbon footprint of a manufactured product;
- C1.4** explain the benefits of developing an environmentally friendly product (*e.g., a windmill that develops energy from wind power*) and assess its potential effectiveness.

C2. Technology and Society

By the end of this course, students will:

- C2.1** explain how the globalization of the manufacturing industry affects the economy of the local community, the province of Ontario, and Canada as a whole;
- C2.2** explain the importance of demographics, geography, and strategic plant location as factors to be considered in setting up a successful manufacturing facility, and describe possible short-term and/or long-term societal implications locally and beyond (*e.g., regional or provincial planning issues, effects on the indigenous population, ecosystem and/or habitat considerations*);
- C2.3** assess the effects of emerging manufacturing technology industries on culture and society from various perspectives (*e.g., safety, technical, financial, business*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with health and safety legislation, standards, and practices, including methods to address deficiencies, as they relate to the manufacturing industry;
- D2.** demonstrate an understanding of the importance of group dynamics and effective leadership in the manufacturing environment;
- D3.** demonstrate an understanding of the postsecondary programs associated with manufacturing, and establish a personalized career pathway.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** demonstrate an understanding of the specific components of legislation and standards related to workplace safety in the manufacturing industry (*e.g., Occupational Health and Safety Act [OHSA], Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (*e.g., Workplace Safety and Insurance Board [WSIB], Industrial Accident Prevention Association [IAPA]*);
- D1.2** describe and assess ways to promote safe and productive work practices in the manufacturing workplace (*e.g., develop and use a safety checklist for work practices, tools, equipment, and operations; develop and conduct safety audits and inspections of the school manufacturing facility; design a plan to address health and safety deficiencies; develop an emergency action plan to implement in the event of a spill*);
- D1.3** demonstrate good housekeeping practices in the work environment (*e.g., cleaning up spills and leaks, keeping areas clean and clear of obstructions, properly organizing tools and equipment*);
- D1.4** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);
- D1.5** use proper ventilation and/or filtration systems to control air quality (*e.g., to minimize the*

effects of welding fumes, plastic off-gassing, cutting-fluid misting, and heat treating);

- D1.6** demonstrate an understanding of the health hazards and injuries associated with workplace conditions (*e.g., repetitive motion and carpal tunnel syndrome, noise and hearing loss, radiation, pinch points*);
- D1.7** explain how ergonomics can affect productivity, product quality, and employee needs and satisfaction, and identify ergonomic considerations related to workshop layout and set-up (*e.g., material handling, ease of movement, lighting, workstation design, organization of tools and equipment*);
- D1.8** use and maintain protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Leadership

By the end of this course, students will:

- D2.1** demonstrate an understanding of their personal strengths and areas for improvement in preparation for a career in the manufacturing industry;
- D2.2** describe the role of management in the manufacturing industry in promoting equity, diversity, and non-discriminatory practices in the workplace;
- D2.3** demonstrate the ability to work in various roles (*e.g., designer, scheduler, technician, supervisor*) as part of a manufacturing team from the concept stage to product completion;

D2.4 describe on the basis of research the requirements for a professional or managerial certification or designation (*e.g., Certified Engineering Technologist, Certified in Production and Inventory Management, Licensed Professional Engineer*) relevant to a career in manufacturing.

D3. Career Opportunities

By the end of this course, students will:

D3.1 demonstrate an understanding of the range of career opportunities within the manufacturing industry (*e.g., management, marketing, finance, purchasing, sales, production, quality control, engineering*);

D3.2 describe postsecondary programs associated with the manufacturing industry and evaluate the appropriateness of these programs with respect to their personal career plans;

D3.3 demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., decision making, job task planning and organizing, problem solving*);

D3.4 demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., teamwork, working independently, organization*);

D3.5 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (*e.g., Passport to Safety certificate, Ontario Skills Passport Work Plan and Transition Plan, photographs of a project, technical drawings, reports, assignments*), and explain why having a current portfolio is important for career development and advancement.